# Solving compound inequalities

**1** Consider the statement $3 < 6x + 7 < 9$:

**a** Write the two inequalities that make up $3 < 6x + 7 < 9$?

**b** Identify the word joining the two inequalities that were obtained from the compound inequality $3 < 6x + 7 < 9$.

**2** A high level of LDL cholesterol ("bad cholesterol") in the blood increases a person's risk of heart disease. The table shows how LDL levels affect risk.
Use $x$ as the variable that represents the LDL cholesterol number to write a single or compound inequality for:

**a** The optimal risk category

**b** The Near optimal / above optimal risk category.

**c** The Very high risk category.

| LDL Cholesterol | Risk Category |
|---|---|
| Less than 100 | Optimal |
| $100 - 129$ | Near optimal / above optimal |
| $130 - 159$ | Borderline high |
| $160 - 189$ | High |
| 190 and above | Very high |

**3** Determine whether the following statements are true for the compound inequality:
$x - 2 < 4$ and $4x > 11$

**a** Both $x = 3$ and $x = 6$ are solutions to the compound inequality.

**b** Neither $x = 3$ nor $x = 6$ are solutions to the compound inequality.

**c** $x = 3$ is a solution to the compound inequality.

**d** $x = 6$ is a solution to the compound inequality.

**4** Determine whether the following values of $x$ are solutions to the compound inequality:
$3 - x > -7$ and $3 - x \leq 6$

**a** $x = -4$ **b** $x = 0$

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**5** The steps to solve $2 < 6x + 8 < 10$ have been jumbled up below. Number the steps from $1$ to $5$.

| Step Number | Step |
|---|---|
| ▯ | Divide each part by $6$. |
| ▯ | Simplify by combining constant terms. |
| ▯ | Subtract $8$ from all three parts. |
| ▯ | Simplify any divisions. |
| $1$ | Write the given compound inequality. |

**6** The formula for converting temperatures from Fahrenheit to Celsius: $C = \frac{5}{9}(F - 32)$.
During a recent year, the temperatures in London, ranged from $-5\ ^\circ\text{C}$ to $35\ ^\circ\text{C}$.
Form a compound inequality to solve for the corresponding range of values of $F$, the temperature in London, in degrees Fahrenheit.

**7** The velocity, in feet per second, of a tennis ball $t$ seconds after it is projected directly upward is $v = 89 - 32t$.
Solve for the range of times $t$ at which the velocity will be between $41\ \text{ft/sec}$ and $73\ \text{ft/sec}$ (not inclusive). Write your answer as an inequality.

**8** To get a final grade of $B$, a student must have an average mark on five tests that is greater than or equal to $80$ and less than $90$.
If Kathleen's grades on the first four tests were $75$, $96$, $97$ and $87$, solve for the range of grades $x$ that she could score on the fifth test that would result in her getting a final grade of $B$. Write your answer as an inequality.

**9** The cost of renting a bounce house includes a fixed charge of $\$325$ for setup and delivery, and a variable charge of $\$175$ for each hour the bounce house is rented for.
Yvonne would like to rent a bounce house for a party, and is willing to spend a minimum amount of $\$1200$ and a maximum amount of $\$1725$. Solve for the range of hours $t$ that she could hire the bounce house for. Write your answer as an inequality.

**10** Ivan scored $88$, $72$, $77$, $79$ and $74$ on his five tests during the semester. The average of those scores count for two-thirds of his final grade. He has a final exam at the end of the semester that will count for the other one-third.
To get a $B$, his final grade must be greater than or equal to $80$ and less than $90$. Solve for the range of grades $x$ that he could score on his final exam that would result in him getting a $B$. Write your answer as an inequality. Assume that $100$ is the highest possible grade.

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**11** Determine whether the following values of $x$ are solutions to the compound inequality.
$2x < -3$ or $3x \geq 4$

**a** $x = 2$ **b** $x = 0$

**12** Determine the solution for each of the following compound inequalities.

**a** $x < 4$ or $x > 5$ **b** $x > 2$ or $x > 4$ **c** $x > 2$ or $x < 5$. **d** $x > 6$ or $x > 3$

**13** For each of the following compound inequalities:

**i** Solve for $x$ in the first inequality.

**ii** Solve for $x$ in the second inequality.

**iii** Write the solution to the compound inequality.

**iv** Plot the solution set on a number line

**a** $x + 5 < -4$ or $x - 4 > 2$.

**b** $5x > -20$ or $-6x < 12$

**c** $\frac{x}{2} \leq 1$ or $7 \geq x - 3$

**d** $x - 5 < 6$ or $x + 5 > 6$

**e** $-5 + x > -4$ or $5x + 1 > 26$

**f** $2x - 18 < -0.4$ or $2x - 18 > 0.4$

## Additional questions

**14** For the following pairs of inequalities, plot the solution to the compound inequality that is Inequality 1 **and** Inequality 2 on a numberline.

**a** Inequality 1: $x \geq 2$
Inequality 2: $x \leq 7$

**b** Inequality 1: $x > 3$
Inequality 2: $x \leq 7$

**15** For the following pairs of inequalities, plot the solution to the compound inequality that is Inequality 1 **or** Inequality 2 on a number line.

**a** Inequality 1: $x \leq 3$
Inequality 2: $x > 8$

**b** Inequality 1: $x < 8$
Inequality 2: $x > 3$

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**16** For the following compound inequalities:

i Rewrite the compound inequality as a pair of inequalities joined by either **and** or **or**.

ii Plot the solution to the compound inequality on a number line.

iii Write the solution in interval notation.

**a** $-5 < x < 8$ **b** $-7 < x \leq 5$

**17** For the following pairs of inequalities, state the solution to the compound inequality that is Inequality 1 **and** Inequality 2.

**a** Inequality 1: $-3x + 2 \leq -13$
Inequality 2: $-4x - 2 \geq 26$

**b** Inequality 1: $4x + 3 \geq -17$
Inequality 2: $3x - 4 < 5$

**18** For the following pairs of inequalities, state the solution to the compound inequality that is Inequality 1 **or** Inequality 2.

**a** Inequality 1: $5x - 7 < -22$
Inequality 2: $-6x + 8 < -28$

**b** Inequality 1: $-6x + 8 < 26$
Inequality 2: $5x - 4 \leq 26$

**19** Carlton tried to solve the following compound inequality, but made a mistake in his work:

$$\begin{aligned}
&\text{Step 0:} && -8 \leq 10 - 3x < 4 \\
&\text{Step 1:} && -18 \leq -3x < -6 \\
&\text{Step 2:} && 6 \leq 3x < 18 \\
&\text{Step 3:} && 2 \leq x < 6
\end{aligned}$$

Determine which step is incorrect and explain the error.

**20** Skyler tried to plot the solution to the compound inequality $4x - 1 \leq 11$ and $\dfrac{x+1}{2} > 5$ on a number line, however his answer is incorrect.




Identify the error(s) and explain how to rectify them.

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